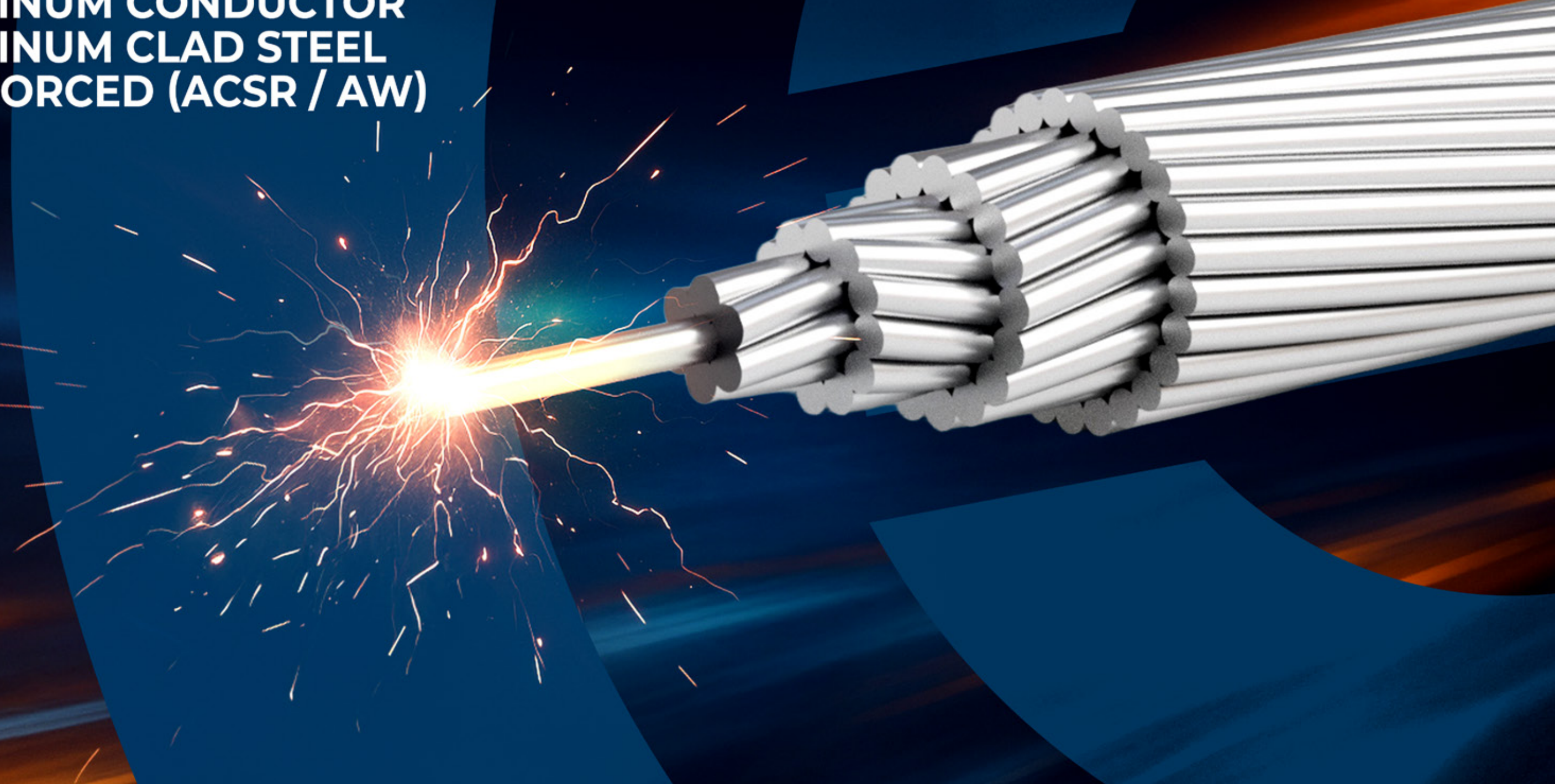


**ALUMINUM CONDUCTOR
ALUMINUM CLAD STEEL
REINFORCED (ACSR / AW)**



Aluminum Conductor Aluminum Clad Steel Reinforced
(ACSR / AW)

Description

An outer layer of Aluminum conductor concentrically stranded over the central core of Aluminum clad Steel form Aluminum Conductor Aluminum Clad Steel Reinforced.

Application

Aluminum Conductors, Aluminum-Clad Steel Reinforced (ACSR/AW) can be used in Medium, High and Extra-High voltage transmission lines; also used for earth wires.

In comparison with ACSR conductors, ACSR/AW conductors have considerable technical and economical advantages in overhead lines.

Its lower weight combined with its higher current carrying capacity and corrosion protection provide a longer life cycle, reduction in energy losses and significant cost saving during the operation of the line.

Technical Data

Relevant Standard	IEC 61089, ASTM B 549
Conductor	Plain bare hard drawn Aluminum - H14 over the central core of Aluminum clad steel solid or stranded wires
Minimum bending Radius	25 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



Product Code	Aluminum Cross Sectional Area		Aluminum Cross Sectional Area		Number and Nominal Diameter of wires		Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Calculated Breaking Load KN	Approx. Weight Kg/km	Rated Current Amps
	kcmil	mm2	kcmil	mm2	Aluminum No. x mm	Steel No. x mm					
A - ASTM B 549											
4 SWAN / AW	41.74	21.20	42.70	21.60	6 x 2.12	1 x 2.12	1.2822	6.35	8.0	81.0	137
4 SWANATE / AW	41.74	21.20	43.00	21.80	7 x 1.96	1 x 2.61	1.2511	6.53	10.0	93.0	140
3 SWALLOW / AW	52.62	26.70	53.90	27.30	6 x 2.38	1 x 2.38	1.0174	7.14	10.0	103.0	159
2 SPARROW / AW	66.36	33.60	67.10	34.00	6 x 2.67	1 x 2.67	0.8084	8.03	12.0	129.0	184
2 SPARATE / AW	66.36	33.60	67.60	34.30	7 x 2.47	1 x 3.3	0.7874	8.26	16.0	149.0	188
GROUSE / AW **	80.00	40.50	82.70	41.90	8 x 2.54	1 x 4.24	0.6359	9.32	22.0	205.0	216
1 ROBIN / AW	83.69	42.40	85.40	43.30	6 x 3	1 x 3	0.6403	8.99	15.0	162.0	213
PETREL / AW **	101.80	51.60	107.80	54.60	12 x 2.34	7 x 2.34	0.4684	11.71	44.0	342.0	269
RAVEN / AW	105.60	63.50	107.70	54.60	6 x 3.37	1 x 3.37	0.5074	10.11	19.0	205.0	248
MINORCA / AW **	110.80	56.10	117.30	59.40	12 x 2.44	7 x 2.44	0.4308	12.22	48.0	372.0	283
2/0 QUAIL / AW	133.10	67.40	135.20	68.50	6 x 3.78	1 x 3.78	0.4033	11.35	23.0	259.0	287
LEGHORN / AW **	134.60	68.20	142.70	72.30	12 x 2.69	7 x 2.69	0.3544	13.46	58.0	452.0	321
GUINEA / AW **	159.00	80.60	168.00	85.10	12 x 2.92	7 x 2.92	0.3008	14.63	68.0	534.0	357
3/0 PIGEON / AW	167.80	85.00	170.70	86.50	6 x 4.25	1 x 4.25	0.3191	12.75	28.0	326.0	333
DOTTEREL / AW **	179.60	89.60	187.10	95.00	12 x 3.08	7 x 3.08	0.2704	15.42	75.0	594.0	382
DORKING / AW **	190.80	96.70	201.90	102.00	12 x 3.2	7 x 3.2	0.2505	16.03	81.0	641.0	402
BRAHMA / AW **	203.20	103.00	220.70	112.00	16 x 2.86	19 x 2.48	0.2163	18.14	121.0	894.0	448
COCHIN / AW **	211.30	107.00	223.00	113.00	12 x 3.37	7 x 3.37	0.2258	16.87	88.0	710.0	429
4/0 PENGUIN / AW	211.60	107.00	215.40	109.00	6 x 4.77	1 x 4.77	0.2533	14.30	34.0	412.0	387
WAXWING / AW	266.80	135.00	268.40	136.00	18 x 3.09	1 x 3.09	0.2096	15.47	30.0	421.0	435
PARTRIDGE / AW	266.80	135.00	271.20	137.00	26 x 2.57	7 x 2	0.2035	16.31	48.0	519.0	448
OSTRISH / AW	300.00	152.00	304.80	154.00	26 x 2.73	7 x 2.12	0.1804	17.27	54.0	583.0	483
MERLIN / AW	336.40	170.00	337.80	171.00	18 x 3.47	1 x 3.47	0.1662	17.37	38.0	531.0	504
LINNET / AW	336.40	170.00	341.30	173.00	26 x 2.89	7 x 2.25	0.1609	18.29	60.0	655.0	520
ORIOLE / AW	336.40	170.00	343.70	174.00	30 x 2.69	7 x 2.69	0.1578	18.82	74.0	737.0	530
CHICKADEE / AW	397.50	201.00	399.20	202.00	18 x 3.77	1 x 3.77	0.1408	18.87	44.0	628.0	561

The above data is approximate and subject to manufacturing tolerances

Note:-

The above values based on Aluminum clad steel (normal strength), different type of Aluminum clad steel core are also according to customer request as per ASTM B 502

** ACSR / AW with High Strength Steel Wires

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

Product Code	Aluminum Cross Sectional Area		Aluminum Cross Sectional Area		Number and Nominal Diameter of wires		Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Calculated Breaking Load KN	Approx. Weight Kg/Km	Rated Current Amps
	kcml	mm2	kcml	mm2	Aluminum No. x mm	Steel No. x mm					
A - ASTM B 549											
BRANT / AW	397.50	201.00	403.00	204.00	24 x 3.27	7 x 2.18	0.1377	19.61	63.0	731.0	574
IBIS / AW	397.50	201.00	403.30	204.00	26 x 3.14	7 x 2.44	0.1363	19.89	70.0	774.0	579
LARK / AW	397.50	201.00	406.00	206.00	30 x 2.92	7 x 2.92	0.1339	20.47	87.0	869.0	589
PELICAN / AW	477.70	242.00	479.60	243.00	18 x 4.14	1 x 4.14	0.1168	20.68	51.0	755.0	632
FLICKER / AW	477.70	242.00	483.00	245.00	24 x 3.58	7 x 2.39	0.1148	21.49	74.0	877.0	645
HAWK / AW	477.70	242.00	484.60	246.00	26 x 3.44	7 x 2.68	0.1136	21.79	84.0	929.0	651
HEN / AW	477.70	242.00	487.90	247.00	30 x 3.2	7 x 3.2	0.1115	22.43	104.0	1043.0	663
OSPREY / AW	556.50	282.00	559.00	283.00	18 x 4.47	1 x 4.47	0.1002	22.33	59.0	880.0	698
PARAKEET / AW	556.50	282.00	564.00	286.00	24 x 3.87	7 x 2.58	0.0983	23.22	86.0	1022.0	713
DOVE / AW	556.50	282.00	564.80	286.00	26 x 3.72	7 x 2.89	0.0971	23.55	97.0	1083.0	720
EAGLE / AW	556.50	282.00	569.70	289.00	30 x 3.46	7 x 3.46	0.0954	24.21	119.0	1217.0	732
PEACOCK / AW	605.00	307.00	612.70	310.00	24 x 4.03	7 x 2.69	0.0906	24.21	93.0	111.0	751
SQUAB / AW	605.00	307.00	614.60	311.00	26 x 3.87	7 x 3.01	0.0898	24.54	105.0	1177.0	757
WOOD DUCK / AW	605.00	307.00	618.80	314.00	30 x 3.61	7 x 3.61	0.0876	25.25	126.0	1323.0	773
TEAL / AW	605.00	307.00	618.40	313.00	30 x 3.61	19 x 2.16	0.0878	23.98	127.0	1314.0	761
KINGBIRD / AW	636.00	322.00	639.40	324.00	18 x 4.78	1 x 4.78	0.0876	23.88	67.0	1006.0	773
SWIFT / AW	636.00	322.00	637.70	323.00	36 x 3.38	1 x 3.38	0.0884	23.62	61.0	946.0	775
ROOK / AW	636.00	322.00	644.00	326.00	24 x 4.14	7 x 2.76	0.0859	24.82	98.0	1168.0	777
GROSBEAK / AW	636.00	322.00	646.10	327.00	26 x 3.97	7 x 3.09	0.0853	25.15	110.0	1238.0	783
SOOTER / AW	636.00	322.00	650.50	330.00	30 x 3.7	7 x 3.7	0.0834	25.88	130.0	1391.0	798
EGRET / AW	636.00	322.00	650.20	329.00	30 x 3.7	19 x 2.22	0.0835	25.88	133.0	1381.0	798
FLAMINGO / AW	666.60	338.00	675.40	342.00	24 x 4.23	7 x 2.82	0.0823	25.40	103.0	1225.0	799
GANNET / AW	666.60	338.00	676.60	343.00	26 x 4.07	7 x 3.16	0.0812	25.76	116.0	1298.0	807
STILT / AW	715.50	363.00	725.00	367.00	24 x 4.39	7 x 2.92	0.0764	26.31	110.0	1314.0	837
STARLING / AW	715.50	363.00	727.40	369.00	26 x 4.21	7 x 3.28	0.0758	26.70	122.0	1393.0	844
REDWING / AW	715.50	363.00	730.90	370.00	30 x 3.92	19 x 2.35	0.0744	27.46	149.0	1552.0	859
COOT / AW	795.00	403.00	797.20	404.00	36 x 3.77	1 x 3.77	0.0711	26.42	74.0	1183.0	868
CUCKOO / AW	795.00	403.00	805.00	408.00	24 x 4.62	7 x 3.08	0.0690	27.74	122.0	1460.0	894
DRAKE / AW	795.00	403.00	807.60	409.00	26 x 4.44	7 x 3.45	0.0682	28.14	136.0	1549.0	903
TERN / AW	795.00	403.00	800.40	406.00	45 x 3.38	7 x 2.25	0.0701	27.00	96.0	1298.0	880
CONDOR / AW	795.00	403.00	805.00	408.00	54 x 3.08	7 x 3.08	0.0690	27.74	124.0	1458.0	894
MALLARY / AW	795.00	403.00	812.70	412.00	30 x 4.14	19 x 2.48	0.0667	28.96	165.0	1726.0	921

RUDDY / AW	795.00	403.00	906.10	459.00	45 x 3.59	7 x 2.4	0.0621	28.73	107.0	1470.0	951
CANARY / AW	900.00	456.00	911.40	462.00	54 x 3.28	7 x 3.28	0.0608	29.51	138.0	1653.0	969
CATBIRD / AW	900.00	456.00	956.60	485.00	36 x 4.14	1 x 4.14	0.0589	28.96	87.0	1420.0	978
RAIL / AW	954.00	483.00	960.40	487.00	45 x 3.7	7 x 2.47	0.0585	29.59	113.0	1558.0	988
CARDINAL / AW	954.00	483.00	966.10	490.00	54 x 3.38	7 x 3.38	0.0573	30.38	146.0	1752.0	1006
TANAGER / AW	1033.50	524.00	1035.80	525.00	36 x 4.3	1 x 4.3	0.0546	30.12	94.0	1537.0	1027
ORTOLAN / AW	1033.50	524.00	1040.00	527.00	45 x 3.85	7 x 2.57	0.0540	30.78	121.0	1688.0	1039
CURLEW / AW	1033.50	524.00	1046.10	530.00	54 x 3.51	7 x 3.51	0.0531	31.62	158.0	1896.0	1056
BLUEJAY / AW	1113.00	564.00	1120.50	568.00	45 x 4	7 x 2.66	0.0501	31.98	130.0	1819.0	1090
FINCH / AW	1113.00	564.00	1127.80	571.00	54 x 3.65	19 x 2.19	0.0494	32.84	167.0	2043.0	1106
BUNTING / AW	1192.50	604.00	1201.00	609.00	45 x 4.14	7 x 2.76	0.0467	33.07	139.0	1948.0	1138
GRACKLE / AW	1192.50	604.00	1206.70	611.00	54 x 3.77	19 x 2.27	0.0463	33.99	179.0	2188.0	1152
SKYLARK / AW	1272.00	645.00	1275.40	646.00	36 x 4.78	1 x 4.78	0.0442	33.43	114.0	1893.0	1172
BITTERN / AW	1272.00	645.00	1280.60	649.00	45 x 4.27	7 x 2.85	0.0439	34.16	149.0	2078.0	1184
PHEASANT / AW	1272.00	645.00	1287.70	652.00	54 x 3.9	19 x 2.34	0.0433	35.10	189.0	2333.0	1201
DIPPER / AW	1351.50	685.00	1360.10	689.00	45 x 4.4	7 x 2.93	0.0414	35.20	158.0	2207.0	1228
MARTIN / AW	1351.50	685.00	1367.70	693.00	54 x 4.02	19 x 2.41	0.0407	36.17	201.0	2478.0	1248
BOBOLINK / AW	1431.00	725.00	1440.20	730.00	45 x 4.53	7 x 3.02	0.0390	36.25	167.0	2336.0	1274
PLOVER / AW	1431.00	725.00	1448.90	734.00	54 x 4.14	19 x 2.48	0.0384	37.21	212.0	2625.0	1294
NUTHATCH / AW	1510.50	765.00	1520.50	770.00	45 x 4.65	7 x 3.1	0.0370	37.24	177.0	2467.0	1317
PARROT / AW	1510.50	765.00	1528.20	774.00	54 x 4.25	19 x 2.55	0.0364	38.23	224.0	2768.0	1337
LAPWING / AW	1590.00	806.00	1601.20	811.00	45 x 4.78	7 x 3.18	0.0351	38.20	186.0	2598.0	1360
FALCON / AW	1590.00	806.00	1609.80	816.00	54 x 4.36	19 x 2.62	0.0346	39.24	236.0	2917.0	1380

The above data is approximate and subject to manufacturing tolerances

Note:-

The above values based on Aluminum clad steel (normal strength), different type of Aluminum clad steel core are also according to customer request as per ASTM B 502

** ACSR / AW with High Strength Steel Wires

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 35 °C , final temperature of 80 °C, wind velocity of 0.6 m/sec, solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

Aluminum Conductor Aluminum Clad Steel Reinforced
(ACSR / AW)



Product Code	Total Sectional Area mm²	Number and Nominal Diameter of wires		Max. D.C. Conductor Resistance at 20 °C Ω/km	Approx. Overall Diameter mm²	Calculated Breaking Load KN	Approx. Weight Kg/km	Rated Current Amps
		Aluminum No. x mm	Aluminum Clad Steel No. x mm					
B - IEC 61089 (A1/SA1A)								
A1/SA1A - 16	17.90	6 x 1.81	1 x 1.81	1.7923	5.43	5.91	59.0	148
A1/SA1A - 25	28.00	6 x 2.26	1 x 2.26	1.1471	6.78	9.00	92.1	197
A1/SA1A - 40	44.80	6 x 2.85	1 x 2.85	0.7169	8.55	14.21	147.4	265
A1/SA1A - 63	70.60	6 x 3.58	1 x 3.58	0.4552	10.70	21.17	232.2	355
A1/SA1A - 100	112.00	6 x 4.51	1 x 4.51	0.2868	13.50	31.84	368.6	478
A1/SA1A - 125	130.00	18 x 2.95	1 x 2.95	0.2304	14.80	29.18	384.3	548
A1/SA1A - 125	140.00	26 x 2.43	7 x 1.89	0.2308	15.40	44.49	460.8	554
A1/SA1A - 160	167.00	18 x 3.34	1 x 3.34	0.1800	16.70	36.38	491.9	642
A1/SA1A - 160	179.00	26 x 2.74	7 x 2.13	0.1803	17.40	56.18	589.8	649
A1/SA1A - 200	208.00	18 x 3.74	1 x 3.74	0.1440	18.70	43.62	614.9	741
A1/SA1A - 200	223.00	26 x 3.07	7 x 2.39	0.1443	19.40	69.27	737.2	749
A1/SA1A - 250	268.00	22 x 3.76	7 x 2.09	0.1153	21.30	67.80	830.9	861
A1/SA1A - 250	279.00	26 x 3.43	7 x 2.67	0.1154	21.70	86.58	921.5	865
A1/SA1A - 315	331.00	45 x 2.96	7 x 1.97	0.0917	23.70	78.33	996.4	995
A1/SA1A - 315	352.00	26 x 3.85	7 x 2.99	0.0916	24.40	107.58	1161.1	1004
A1/SA1A - 400	420.00	45 x 3.34	7 x 2.22	0.0722	26.70	97.50	1265.3	1160
A1/SA1A - 400	438.00	54 x 3.02	7 x 3.02	0.0723	27.20	124.20	1402.9	1166
A1/SA1A - 450	473.00	45 x 3.54	7 x 2.36	0.0642	28.30	107.48	1423.4	1250
A1/SA1A - 450	492.00	54 x 3.21	7 x 3.21	0.0642	28.90	139.72	1578.2	1258
A1/SA1A - 500	525.00	45 x 3.73	7 x 2.49	0.0578	29.80	119.42	1581.6	1336
A1/SA1A - 500	547.00	54 x 3.38	7 x 3.38	0.0578	30.40	153.99	1753.6	1344
A1/SA1A - 560	589.00	45 x 3.95	7 x 2.63	0.0516	31.60	133.75	1771.4	1437
A1/SA1A - 560	612.00	54 x 3.58	19 x 2.15	0.0516	32.20	169.36	1956.3	1445

Al/SA1A - 630	662.00	45 x 4.19	7 x 2.79	0.0458	33.50	150.47	1992.8	1548
Al/SA1A - 630	688.00	54 x 3.79	19 x 2.28	0.0459	34.20	190.52	2200.9	1557
Al/SA1A - 710	746.00	45 x 4.44	7 x 2.96	0.0407	35.60	169.57	2245.8	1668
Al/SA1A - 710	775.00	54 x 4.03	19 x 2.42	0.0407	36.30	214.72	2480.3	1678
Al/SA1A - 800	874.00	54 x 4.28	19 x 2.57	0.0361	38.50	241.94	2794.7	1807

The above data is approximate and subject to manufacturing tolerances

Note:-

The above values based on Aluminum clad steel 20SA-Type A (regular tesnile strength), different type of Aluminum clad steel core are also according to customer request as per IEC 61232

The above values of current carrying capacity are calculated due to frequency up to 60 Hz, assuming an ambient temperature of 20 °C , final temperature of 80 °C, wind velocity of 1.0 m/sec.

solar radiation of 900 W/m2, emissivity coefficient of 0.6, solar radiation absorption coefficient of 0.5

